When two samples of matter that are initially at different temperatures are placed in thermal contact, the samples undergo heat transfer until reaching thermal equilibrium at some new, final temperature. Transferred heat ($q$) can be measured based on the initial and final temperatures ($T_{initial}$ and $T_{final}$) of a sample that is placed into a known volume of water inside the insulated chamber of a calorimeter.

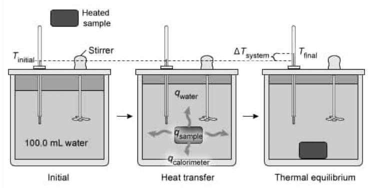



**Figure 1** Analysis of a heated sample using a calorimeter

According to the first law of thermodynamics, the transferred heat energy is conserved, and the heat lost from the sample (indicated by a negative sign) must be equal to the sum of the heat gained by the water and the calorimeter, as expressed by Equation 1.

$$-q_{sample} = q_{water} + q_{calorimeter}$$

**Equation 1**

In a perfectly efficient system, $q_{calorimeter} = 0$, and all the heat released from the sample is retained by the water. In real systems, some heat is absorbed by the calorimeter itself. Based on the amount of heat transferred from the sample and the change in temperature, the heat capacity ($C$) of the entire sample can be determined. Expressing the heat capacity per unit mass yields the specific heat capacity ($C_p$) of the substance comprising the sample.

**Table 1** Measured specific heat capacities of several selected substances

| Sample | Specific heat capacity (J/g·°C) |
|---|---|
| Lead | 0.129 |
| Tungsten | 0.132 |
| Silver | 0.235 |
| Strontium | 0.315 |
| Zinc | 0.388 |
| Cobalt | 0.421 |
| Titanium | 0.525 |
| Wood | 2.00 |
| Paraffin wax | 2.5 |
| Water | 4.184 |

Experiment 1

A sample of water (10.0 mL at 75 °C) was stirred into a calorimeter containing 100.0 mL of water initially at 21 °C. At thermal equilibrium, the system was found to have a temperature of 25 °C.

Experiment 2

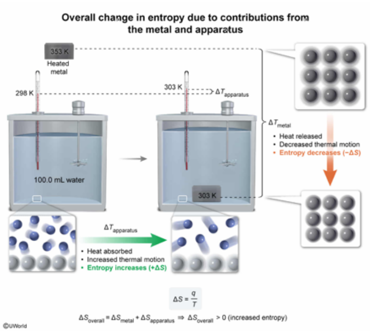
The water in the calorimeter from Experiment 1 was discarded and replaced with 100.0 mL of fresh water at 25 °C. An unidentified 100.0 g sample of one of the metals from Table 1 was heated to an initial temperature of 80 °C and then placed into the water. At thermal equilibrium, the system was found to have a temperature of 30 °C.

Based on the results of Experiment 2, which of the following statements is true concerning the metal sample and the water-calorimeter system (apparatus)?

- ○ A. Both the metal sample and the apparatus experience an increase in entropy, resulting in an overall increase in entropy within the system. [12%]
- ✔ ◉ B. The entropy of the metal sample decreases, but the entropy of the apparatus increases by a greater amount, resulting in an overall increase in entropy within the system. [73%]
- ○ C. The entropy of the metal sample increases, but the entropy of the apparatus decreases by a greater amount, resulting in an overall decrease in entropy within the system. [8%]
- ○ D. Both the metal sample and the apparatus experience a decrease in entropy, resulting in an overall decrease in entropy within the system. [4%]

Correct
73% answered correctly

**Explanation:**




According to the **second law of thermodynamics**, the total entropy of an isolated system will always increase over time during any spontaneous process.  Although entropy indicates the disorder of a system, entropy is more fundamentally a measure of how much energy in the system is unavailable to do work.  As such, the instantaneous change in entropy $\Delta S$ for a process is equal to the ratio of the transferred heat $q$ to the absolute (Kelvin) temperature $T$ at which the transfer occurs.

$$\Delta S = \frac{q}{T}$$

In any thermal energy transfer, some of the energy of the system becomes distributed in the random, **thermal motion** of the atoms and molecules.  As the disorder increases, part of the energy is made unavailable as **disbursed heat** and cannot be converted to work.

During Experiment 2, the **loss of heat** by the metal sample **decreases the thermal motion** (ie, vibrations) of its atoms and **lowers the entropy** of the sample.  The $\Delta S$ can be estimated using the average temperature

$$T_{\text{average}} = \frac{T_{\text{final}} + T_{\text{initial}}}{2}$$

from the $\Delta T$ for the process (converted to Kelvin).

$$T_{\text{average(metal)}} = \frac{(30\,°\text{C} + 273) + (80\,°\text{C} + 273)}{2} = \frac{303\text{ K} + 353\text{ K}}{2} = 328\text{ K}$$

$$\Delta S_{\text{metal}} = \frac{-q_{\text{metal}}}{T_{\text{average(metal)}}} \approx \frac{-q_{\text{metal}}}{328\text{ K}} \Longrightarrow \text{smaller negative number}$$

Conversely, the **heat absorbed** by the water-calorimeter apparatus (ie, $q_{\text{apparatus}} = q_{\text{water}} + q_{\text{calorimeter}}$) **increases the thermal motion** of the water molecules and **raises the entropy** of the apparatus.

$$T_{\text{average(apparatus)}} = \frac{(30\,°\text{C} + 273) + (25\,°\text{C} + 273)}{2} = \frac{303\text{ K} + 298\text{ K}}{2} = 301\text{ K}$$

$$\Delta S_{\text{apparatus}} = \frac{q_{\text{apparatus}}}{T_{\text{average(apparatus)}}} \approx \frac{+q_{\text{apparatus}}}{301\text{ K}} \Longrightarrow \text{larger positive number}$$

Because the heat loss from the metal occurs at a higher average temperature than the heat gain by the water and calorimeter (the apparatus), the decrease in the entropy of the metal sample is overcome by the increase in the entropy of the apparatus, resulting in an **overall increase in entropy** within the system.

$$\left|\frac{q_{\text{metal}}}{328\text{ K}}\right| < \left|\frac{q_{\text{apparatus}}}{303\text{ K}}\right|$$

$$\Delta S_{\text{overall}} = \Delta S_{\text{metal}} + \Delta S_{\text{apparatus}} \Longrightarrow \Delta S_{\text{overall}} > 0\,(\text{increased entropy})$$

**(Choice A)**  Cooling an object (eg, the metal sample) decreases the thermal motion of its constituent atoms and *decreases* its entropy.

**(Choices C and D)**  The transfer of heat from a warmer object (eg, a heated metal sample) to a cooler object (eg, the apparatus) is a spontaneous process.  According to the second law of thermodynamics, the total entropy of the system will *increase*.

**Educational objective:**
According to the second law of thermodynamics, the total entropy of an isolated system will increase during any spontaneous process. Entropy assesses disorder and is a measure of the extent to which the energy of the system is unavailable to do work.

**Time spent:**0 sec
**Subject:**
General Chemistry

**QID**:403700
**System:**
Thermodynamics, Kinetics & Gas Laws

When two samples of matter that are initially at different temperatures are placed in thermal contact, the samples undergo heat transfer until reaching thermal equilibrium at some new, final temperature. Transferred heat ($q$) can be measured based on the initial and final temperatures ($T_{initial}$ and $T_{final}$) of a sample that is placed into a known volume of water inside the insulated chamber of a calorimeter.

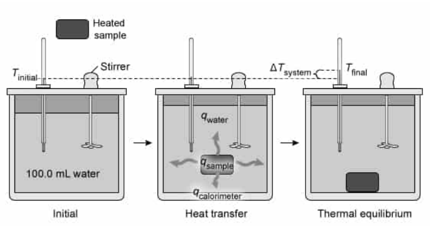



**Figure 1** Analysis of a heated sample using a calorimeter

According to the first law of thermodynamics, the transferred heat energy is conserved, and the heat lost from the sample (indicated by a negative sign) must be equal to the sum of the heat gained by the water and the calorimeter, as expressed by Equation 1.

$$-q_{sample} = q_{water} + q_{calorimeter}$$

**Equation 1**

In a perfectly efficient system, $q_{calorimeter} = 0$, and all the heat released from the sample is retained by the water. In real systems, some heat is absorbed by the calorimeter itself. Based on the amount of heat transferred from the sample and the change in temperature, the heat capacity ($C$) of the entire sample can be determined. Expressing the heat capacity per unit mass yields the specific heat capacity ($C_p$) of the substance comprising the sample.

**Table 1** Measured specific heat capacities of several selected substances

| Sample | Specific heat capacity (J/g·°C) |
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| Water | 4.184 |

Experiment 1

A sample of water (10.0 mL at 75 °C) was stirred into a calorimeter containing 100.0 mL of water initially at 21 °C. At thermal equilibrium, the system was found to have a temperature of 25 °C.

Experiment 2

The water in the calorimeter from Experiment 1 was discarded and replaced with 100.0 mL of fresh water at 25 °C. An unidentified 100.0 g sample of one of the metals from Table 1 was heated to an initial temperature of 80 °C and then placed into the water. At thermal equilibrium, the system was found to have a temperature of 30 °C.

Approximately how much heat energy is gained by the water in the calorimeter during Experiment 2 ?

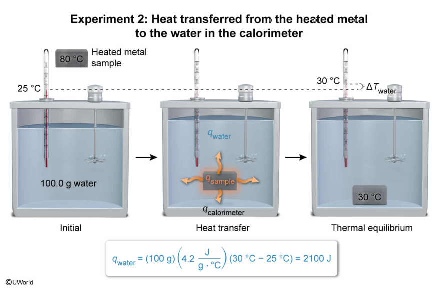
- ○ A. 21 J [8%]
- ○ B. 168 J [10%]

○ C. 1680 J [12%]

✔ ◉ D. 2100 J [67%]

Correct
68% answered correctly

**Explanation:**




The specific heat capacity $C_p$ of a substance (sometimes abbreviated as specific heat) is the amount of heat required to raise the temperature of 1 gram of the substance by 1 °C. Accordingly, if the mass *m* and change in temperature Δ*T* of a substance are known, the $C_p$ of the substance can be used to calculate the heat *q* that is absorbed or released by the substance using the relationship:

$$q = mC_p(T_{final} - T_{initial}) = mC_p\,\Delta T$$

Although the **density of water** decreases slightly as its temperature increases, the density remains very **close to 1.0 g/mL between 1 °C and 100 °C**. As such, the mass of the 100.0 mL of water in the calorimeter can be estimated as:

$$100.0\ \cancel{\text{mL water}} \times \frac{1.0\text{ g water}}{1\ \cancel{\text{mL water}}} = 100.0\text{ g water}$$

At thermal equilibrium, the final temperature of the water is equal to the temperature of the system. Therefore, using the specific heat capacity of water from Table 1 (rounded for estimation purposes), the **heat gained by the water** in Experiment 2 is approximately:

$$q_{water} = (100\text{ g})\left(4.2\frac{\text{J}}{\text{g}\cdot°\text{C}}\right)(30\ °\text{C} - 25\ °\text{C})$$

$$q_{water} = 100\left(4.2\frac{\text{J}}{\cancel{°\text{C}}}\right)(5\ \cancel{°\text{C}}) = 2100\text{ J}$$

Note that it is not necessary here to convert the temperatures to Kelvin because a difference in temperatures is being used, and the difference in degrees Celsius is equivalent to the difference in Kelvin (ie, 30 °C − 25 °C = 303 K − 298 K = 5 °C or K).

**(Choice A)** 21 J is the amount of heat gained by 1 mL (1.0 g) of water, but the total sample is 100 mL (100 g).

**(Choice B)** 168 J is the amount of heat that would be gained by 10 mL (10 g) of water using the temperature change from Experiment 1.

**(Choice C)** 1680 J is the amount of heat gained by the 100 mL (100 g) of water in *Experiment 1*.

**Educational objective:**
The specific heat capacity of a substance is the amount of heat that must be absorbed (or released) to change the temperature of 1 gram of the substance by 1 °C. Multiplying the specific heat capacity by the mass and change in temperature of a sample yields the associated amount of heat absorbed (or released) by the sample.

| | |
|---|---|
| **Time spent:**0 sec | **QID**:403701 |
| **Subject:** | **System:** |
| General Chemistry | Thermodynamics, Kinetics & Gas Laws |

When two samples of matter that are initially at different temperatures are placed in thermal contact, the samples undergo heat transfer until reaching thermal equilibrium at some new, final temperature. Transferred heat ($q$) can be measured based on the initial and final temperatures ($T_{\text{initial}}$ and $T_{\text{final}}$) of a sample that is placed into a known volume of water inside the insulated chamber of a calorimeter.

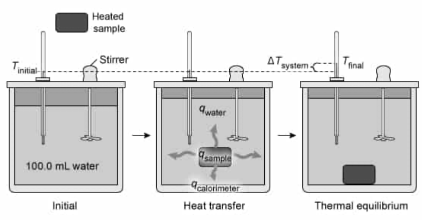



**Figure 1** Analysis of a heated sample using a calorimeter

According to the first law of thermodynamics, the transferred heat energy is conserved, and the heat lost from the sample (indicated by a negative sign) must be equal to the sum of the heat gained by the water and the calorimeter, as expressed by Equation 1.

$$-q_{\text{sample}} = q_{\text{water}} + q_{\text{calorimeter}}$$

**Equation 1**

In a perfectly efficient system, $q_{\text{calorimeter}} = 0$, and all the heat released from the sample is retained by the water. In real systems, some heat is absorbed by the calorimeter itself. Based on the amount of heat transferred from the sample and the change in temperature, the heat capacity ($C$) of the entire sample can be determined. Expressing the heat capacity per unit mass yields the specific heat capacity ($C_p$) of the substance comprising the sample.

**Table 1** Measured specific heat capacities of several selected substances

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| Lead | 0.129 |
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| Titanium | 0.525 |
| Wood | 2.00 |
| Paraffin wax | 2.5 |
| Water | 4.184 |

Experiment 1

A sample of water (10.0 mL at 75 °C) was stirred into a calorimeter containing 100.0 mL of water initially at 21 °C. At thermal equilibrium, the system was found to have a temperature of 25 °C.

Experiment 2

The water in the calorimeter from Experiment 1 was discarded and replaced with 100.0 mL of fresh water at 25 °C. An unidentified 100.0 g sample of one of the metals from Table 1 was heated to an initial temperature of 80 °C and then placed into the water. At thermal equilibrium, the system was found to have a temperature of 30 °C.

Which of the following expressions gives the heat capacity of the calorimeter vessel as determined by Experiment 1?

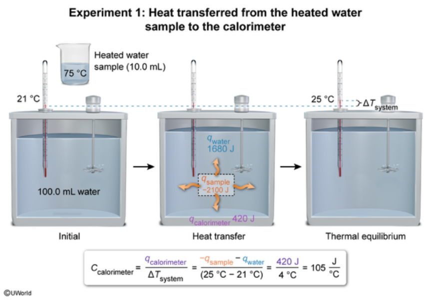
- A. $C_{\text{calorimeter}} = \dfrac{-q_{\text{sample}} - q_{\text{water}}}{m_{\text{water}}} = 1.05\ \text{J/°C}$ [12%]
- B. $C_{\text{calorimeter}} = \dfrac{-q_{\text{sample}}}{\Delta T} = 42\ \text{J/°C}$ [10%]

C. $C_{\text{calorimeter}} = \dfrac{-q_{\text{sample}} - q_{\text{water}}}{\Delta T} = 105\ \text{J}/°\text{C}$ [64%]

D. $C_{\text{calorimeter}} = \dfrac{-q_{\text{water}}}{\Delta T} = 420\ \text{J}/°\text{C}$ [12%]

Correct
63% answered correctly

**Explanation:**




The **heat capacity** *C* of a sample (or object) is the amount of heat required to raise the temperature of the entire sample by 1 °C. In contrast, the specific heat capacity $C_p$ of a substance is the amount of heat required to raise the temperature of *1 gram* of the substance by 1 °C. Consequently, **two objects** that are made of the **same substance** but have **different masses** will each have a **different heat capacity** but an **identical specific heat capacity**.

In Experiments 1 and 2, the mass and composition of the calorimeter are not known. As such, it is more convenient to consider the calorimeter as a single object and determine the heat capacity of the entire apparatus. Rearranging Equation 1 shows that the heat absorbed by the calorimeter is calculated as:

$$q_{\text{calorimeter}} = -q_{\text{sample}} - q_{\text{water}}$$

Multiplying the specific heat capacity of water (from Table 1) by the mass and change in temperature of the heated water sample in Experiment 1 yields the associated amount of **heat released from the heated water sample**.

$$q_{\text{sample}} = mC_p(T_{\text{final}} - T_{\text{initial}}) = \left(10.0\ \cancel{\text{mL}} \times \frac{1.0\ \text{g}}{1\ \cancel{\text{mL}}}\right)\left(4.2\ \frac{\text{J}}{\text{g}\cdot°\text{C}}\right)(25\ °\text{C} - 75\ °\text{C}) = -2100\ \text{J}$$

Likewise, the amount of **heat gained by the water in the calorimeter** is given by:

$$q_{\text{water}} = \text{m}C_p(T_{\text{final}} - T_{\text{initial}}) = \left(100.0\ \cancel{\text{mL}} \times \frac{1.0\ \text{g}}{1\ \cancel{\text{mL}}}\right)\left(4.2\ \frac{\text{J}}{\text{g}\cdot°\text{C}}\right)(25\ °\text{C} - 21\ °\text{C}) = +1680\ \text{J}$$

The **heat absorbed by the calorimeter** is equal to its heat capacity $C_{\text{calorimeter}}$ multiplied by its change in temperature.

$$q_{\text{calorimeter}} = C_{\text{calorimeter}}\ \Delta T$$

Substituting this expression for $q_{\text{calorimeter}}$ into the heat balance (Equation 1) and then solving for $C_{\text{calorimeter}}$ gives:

$$q_{\text{calorimeter}} = -q_{\text{sample}} - q_{\text{water}}$$

$$C_{\text{calorimeter}}\ \Delta T = -q_{\text{sample}} - q_{\text{water}}$$

$$C_{\text{calorimeter}} = \frac{-q_{\text{sample}} - q_{\text{water}}}{\Delta T}$$

At thermal equilibrium, the final temperature of the calorimeter is equal to the temperature of the system. Therefore, substituting the heat and temperature values into the equation above gives the value of $C_{\text{calorimeter}}$:

$$C_{\text{calorimeter}} = \frac{-q_{\text{sample}} - q_{\text{water}}}{\Delta T} = \frac{-(-2100\ \text{J}) - (1680\ \text{J})}{(25\ °\text{C} - 21\ °\text{C})} = \frac{420\ \text{J}}{4\ °\text{C}} = 105\frac{\text{J}}{°\text{C}}$$

**(Choice A)** This expression incorrectly evaluates *specific* heat capacity (ie, the heat capacity per unit of mass) rather than its heat capacity. Calculating the specific heat capacity of the calorimeter (not asked by the question) would require dividing the heat absorbed by the calorimeter by $\Delta T$ and by the mass of the *calorimeter system*, not by the mass of the water initially present in the calorimeter.

**(Choices B and D)** These expressions divide the incorrect quantities of heat by the change in temperature. The heat capacity of the calorimeter is evaluated according to how much heat is absorbed by the calorimeter (ie, the *difference* between how much heat is released from the sample and how much heat is absorbed by the water inside the calorimeter).

**Educational objective:**
The heat capacity of a sample is the amount of heat required to raise the temperature of the entire sample by 1 °C, whereas the specific heat capacity (specific heat) of a substance is the amount of heat required to raise the temperature of 1 gram of a substance by 1 °C.

**Time spent:**0 sec
**Subject:**
General Chemistry

**QID**:403702
**System:**
Thermodynamics, Kinetics & Gas Laws

When two samples of matter that are initially at different temperatures are placed in thermal contact, the samples undergo heat transfer until reaching thermal equilibrium at some new, final temperature. Transferred heat ($q$) can be measured based on the initial and final temperatures ($T_{initial}$ and $T_{final}$) of a sample that is placed into a known volume of water inside the insulated chamber of a calorimeter.

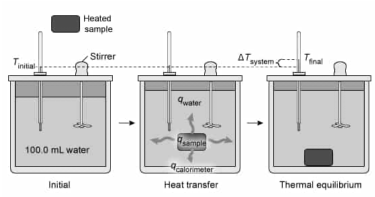



**Figure 1** Analysis of a heated sample using a calorimeter

According to the first law of thermodynamics, the transferred heat energy is conserved, and the heat lost from the sample (indicated by a negative sign) must be equal to the sum of the heat gained by the water and the calorimeter, as expressed by Equation 1.

$$-q_{sample} = q_{water} + q_{calorimeter}$$

**Equation 1**

In a perfectly efficient system, $q_{calorimeter} = 0$, and all the heat released from the sample is retained by the water. In real systems, some heat is absorbed by the calorimeter itself. Based on the amount of heat transferred from the sample and the change in temperature, the heat capacity ($C$) of the entire sample can be determined. Expressing the heat capacity per unit mass yields the specific heat capacity ($C_p$) of the substance comprising the sample.

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| Titanium | 0.525 |
| Wood | 2.00 |
| Paraffin wax | 2.5 |
| Water | 4.184 |

Experiment 1

A sample of water (10.0 mL at 75 °C) was stirred into a calorimeter containing 100.0 mL of water initially at 21 °C. At thermal equilibrium, the system was found to have a temperature of 25 °C.

Experiment 2

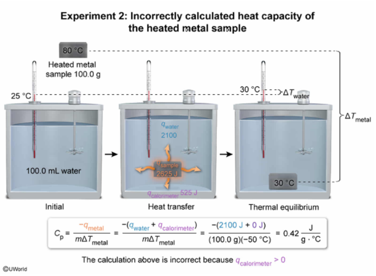
The water in the calorimeter from Experiment 1 was discarded and replaced with 100.0 mL of fresh water at 25 °C. An unidentified 100.0 g sample of one of the metals from Table 1 was heated to an initial temperature of 80 °C and then placed into the water. At thermal equilibrium, the system was found to have a temperature of 30 °C.

Based on the results of the calorimetry experiments, a lab technician concludes that the specific heat capacity of the unidentified metal sample is 0.42 J/g·°C. Is this conclusion correct?

- ○ A. Yes; in Experiment 2 the amount of heat gained by the water is essentially equal to the total amount of heat released by the metal sample because Experiment 1 shows that almost no heat is lost from the water and absorbed by the calorimeter ($q_{calorimeter} \approx 0$). [23%]
- ○ B. Yes; in Experiment 2 the amount of heat gained by the water is less than the total amount of heat released by the metal sample because Experiment 1 shows that the sample absorbs a small amount of heat from the calorimeter ($q_{calorimeter} < 0$). [18%]
- ✔ ◉ C. No; in Experiment 2 the amount of heat gained by the water is less than the total amount of heat released by the metal sample because Experiment 1 shows that a significant amount of heat from a sample is lost from the water and absorbed by the calorimeter ($q_{calorimeter} > 0$). [48%]
- ○ D. No; in Experiment 2 the amount of heat gained by the water is greater than the total amount of heat released by the metal sample because Experiment 1 shows that the calorimeter adds a small amount of heat to the water ($q_{calorimeter} < 0$). [8%]

Correct
48% answered correctly

**Explanation:**




According to the **first law of thermodynamics**, the transferred **heat energy *q* is conserved**, and the heat lost from the sample must be equal to the sum of the heat gained by the water and the calorimeter.

$$-q_{\text{sample}} = q_{\text{water}} + q_{\text{calorimeter}}$$

In a **perfectly efficient** system, $\boldsymbol{q_{\text{calorimeter}} = 0}$ (ie, no heat would be absorbed by the calorimeter), and all the heat released from the sample would be retained by the water.  In reality, **no calorimeter is perfect**, and some heat will always be absorbed by the calorimeter itself because its heat capacity is not zero.

In the given scenario, the lab technician calculates the amount of heat retained by the water in Experiment 2.

$$q_{\text{water}} = mC_{\text{p}}(T_{\text{final}} - T_{\text{initial}}) = \left(100.0\ \cancel{\text{mL}} \times \frac{1.0\ \text{g}}{1\ \cancel{\text{mL}}}\right)\left(4.2\ \frac{\text{J}}{\text{g} \cdot °\text{C}}\right)(30\ °\text{C} - 25\ °\text{C}) = 2100\ \text{J}$$

The lab technician then wrongly assumes that this amount of heat represents all the heat released by the metal and incorrectly calculates the specific heat capacity $C_p$ of the unidentified metal sample.

$$q_{\text{metal}} = mC_{\text{p}}(T_{\text{final}} - T_{\text{initial}})$$

$$-2100\ \text{J} = (100.0\ \text{g})C_{\text{p}}(30\ °\text{C} - 80\ °\text{C})$$

$$C_{\text{p}} = \frac{-2100\ \text{J}}{(100.0\ \text{g})(-50\ °\text{C})} = 0.42 \frac{\text{J}}{\text{g} \cdot °\text{C}}$$

The essential error is that a **significant portion of the heat** released by the metal sample was **absorbed by the calorimeter**.  As such, the total amount of heat retained by the water was less than the total amount released by the metal sample (ie, $\boldsymbol{q_{\text{calorimeter}} > 0}$), and the amount of heat absorbed by the calorimeter is not negligible as assumed in the calculation.

**(Choice A)**  The results of Experiment 1 prove that a significant amount of heat released by a sample is absorbed by the calorimeter (ie, its heat capacity is greater than 0).  Therefore, $q_{\text{calorimeter}}$ is *greater than* 0, and the amount of heat absorbed by the water cannot be equal to the total amount of heat released by the metal sample.

**(Choices B and D)**  Heat transfer occurs from hotter objects to cooler objects.  In the experiments, the samples are hotter than the calorimeter and the water in the calorimeter.  Therefore, the samples and the water cannot absorb a small amount of heat from the calorimeter (ie, $q_{\text{calorimeter}}$ cannot be less than 0).

**Educational objective:**
Heat transfer occurs from hotter objects to cooler objects.  According to the first law of thermodynamics, heat energy is conserved, and the heat lost from a sample must be equal to the sum of the heat gained by substances in thermal contact with the sample.

**Time spent:**0 sec
**Subject:**
General Chemistry

**QID:**403703
**System:**
Thermodynamics, Kinetics & Gas Laws

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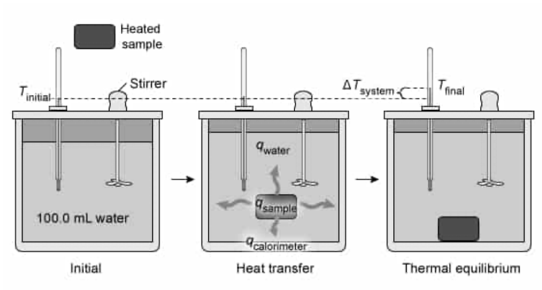


**Figure 1** Analysis of a heated sample using a calorimeter

According to the first law of thermodynamics, the transferred heat energy is conserved, and the heat lost from the sample (indicated by a negative sign) must be equal to the sum of the heat gained by the water and the calorimeter, as expressed by Equation 1.

$$-q_{\text{sample}} = q_{\text{water}} + q_{\text{calorimeter}}$$

**Equation 1**

In a perfectly efficient system, $q_{\text{calorimeter}} = 0$, and all the heat released from the sample is retained by the water. In real systems, some heat is absorbed by the calorimeter itself. Based on the amount of heat transferred from the sample and the change in temperature, the heat capacity ($C$) of the entire sample can be determined. Expressing the heat capacity per unit mass yields the specific heat capacity ($C_p$) of the substance comprising the sample.

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Experiment 1

A sample of water (10.0 mL at 75 °C) was stirred into a calorimeter containing 100.0 mL of water initially at 21 °C. At thermal equilibrium, the system was found to have a temperature of 25 °C.

Experiment 2

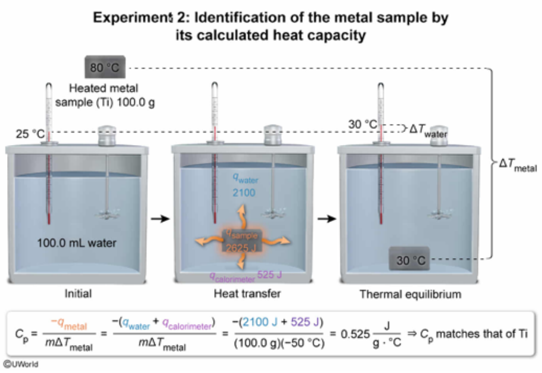
The water in the calorimeter from Experiment 1 was discarded and replaced with 100.0 mL of fresh water at 25 °C. An unidentified 100.0 g sample of one of the metals from Table 1 was heated to an initial temperature of 80 °C and then placed into the water. At thermal equilibrium, the system was found to have a temperature of 30 °C.

According to the results from Experiments 1 and 2, the unidentified metal sample is:

- A. Sr, because $q_{calorimeter} > 0$, which means that less than 2100 J of heat is released by the metal. [17%]
- B. Co, because $q_{calorimeter} \approx 0$, which means exactly 2100 J of heat is released by the metal. [24%]
- ✔ C. Ti, because $q_{calorimeter} > 0$, which means that more than 2100 J of heat is released by the metal. [51%]
- D. unknown and its composition cannot be determined without more information. [6%]

Correct
50% answered correctly

**Explanation:**




According to the **first law of thermodynamics**, the heat energy transferred from the unidentified metal sample is conserved and must be equal to the sum of the heat gained by the water and the calorimeter.

$$-q_{\text{metal}} = q_{\text{water}} + q_{\text{calorimeter}}$$

In Experiment 2, $q_{water}$ is:

$$q_{\text{water}} = mC_{\text{p}}(T_{\text{final}} - T_{\text{initial}}) = \left(100.0\ \cancel{\text{mL}} \times \frac{1.0\ \text{g}}{1\ \cancel{\text{mL}}}\right)\left(4.2\ \frac{\text{J}}{\text{g} \cdot °\text{C}}\right)(30\ °\text{C} - 25\ °\text{C}) = 2100\ \text{J}$$

If the calorimeter were perfectly efficient, then $q_{calorimeter}$ would equal 0, and $-q_{metal}$ would equal

$q_{water}$. Calculating the specific heat capacity $C_p$ for the metal on this assumption gives:

$$q_{metal} = mC_p(T_{final} - T_{initial})$$

$$-2100\ \text{J} = (100.0\ \text{g})C_p(30\ °\text{C} - 80\ °\text{C})$$

$$C_p = \frac{-2100\ \text{J}}{(100.0\ \text{g})(-50\ °\text{C})} = 0.42\frac{\text{J}}{\text{g} \cdot °\text{C}}$$

However, this value of $C_p$ is incorrect (ie, too low) because Experiment 1 shows that $q_{calorimeter}$ is not equal to 0. Consequently, **more than 2100 J is released by the metal** sample (ie, there is an additional amount absorbed by the calorimeter), and the $C_p$ for the metal must be greater than 0.42 J/g·°C. Although the $C_p$ value can be calculated by first determining $C_{calorimeter}$ from Experiment 1 and then evaluating $q_{calorimeter}$, $q_{water}$, and $q_{metal}$ from Experiment 2, this is not necessary to answer the question. Among the metal choices given from Table 1, only titanium (Ti) has a $C_p$ > 0.42 J/g·°C.

**(Choice A)** A value of $C_p$ = 0.315 J/g·°C (matching that of strontium) is obtained by subtracting $q_{calorimeter}$ from $q_{water}$ instead of adding these contributions together in the heat balance step of the calculation.

**(Choice B)** A value of $C_p$ = 0.42 J/g·°C (matching that of cobalt) is obtained by incorrectly assuming that $C_{calorimeter}$ is very small and $q_{calorimeter}$ is negligible.

**(Choice D)** The given information from Experiments 1 and 2 is enough to solve for the heat capacity of the unidentified metal sample and identify a likely match from Table 1.

**Educational objective:**
According to the first law of thermodynamics, heat energy is conserved, and the heat lost from a sample must be equal to the sum of the heat gained by substances in thermal contact with the sample. The specific heat capacity of a substance is the amount of heat required to raise the temperature of 1 gram of the substance by 1 °C.

**Time spent:**0 sec
**Subject:**
General Chemistry

**QID**:403704
**System:**
Thermodynamics, Kinetics & Gas Laws